

CSIT305 – Emerging Information Technologies and their Applications

ShadeShift

Smart Urban Heat-Safety and Adaptive Cooling Network

Group name: GRPLESS

Team Members

Name	Student ID	Work ratio/Contribution
TAN HUI QI CHERI	8963903	100% / Part 1- 2.2
WHYE LI HENG DOMINIC	9891092	100% / Part 2.3 - 3.3
YAN YUXUAN	9071453	100% / Part 3.4 - 4
NEO YI QUAN	8931586	100% / Part 5
CLEON LIM KANG WUE	9533941	100% / Part 6 - 6.4
LI ZHENG XIAO	9088945	100% / Part 6.5
CEDRIC HENG CHENG YEE	9091993	100% / Part 7 and 8

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1. System Overview Description

1.1 Proposed System

Our proposed system would be called **ShadeShift: Smart Urban Heat-Safety and Adaptive Cooling Network**. It is an IoT system based in the cloud which is designed to monitor heat conditions at street level, predict areas that may have dangerous levels of heat exposure, suggest heat-safe routes for pedestrians and **support the automated control of cooling facilities located in public spaces**.

ShadeShift integrates artificial intelligence, edge-computing gateways and cloud services with environmental sensors and geographic information systems. Data could be collected from streets, bus stops, sheltered walkways, parks and bus interchanges. These data would be processed continuously so that more detailed insights on how different locations would have different heat conditions could be discovered.

Unlike conventional weather applications, which typically provide general conditions for a larger geographical area, ShadeShift focuses on heat exposure at the level of individual pedestrian routes and public spaces. Rather than simply informing users that conditions are hot, the system converts environmental data into actionable recommendations, such as selecting a route with lower predicted heat exposure or activating cooling infrastructure in high-risk areas.

1.2 System Goal

The overall goal of ShadeShift is to minimise pedestrians' exposure to dangerous urban heat by providing a hyperlocal heat monitoring system, predicting heat risk, suggesting heat safe routes as well as automating cooling responses.

Singapore has high temperatures and humidity all year round. The Urban Heat Island effect in a dense urban setting may intensify exposure to heat. Urban surfaces such as roads, pavements and buildings then absorb and hold heat leading to built-up areas being warmer than areas with more greenery. Singapore's climate projections also indicate that urban heat and heat stress will increase, making long-term heat-resilience planning more crucial.

The recent climate records of Singapore show the seriousness of the issue. 2024 was Singapore's hottest year on record, said the National Environment Agency. The current definition of high heat stress in Singapore is based on the Wet Bulb Globe Temperature (WBGT). High heat stress is defined as an hourly average WBGT of 33°C and above.

Air temperature is not the only cause of heat exposure. Factors influencing the amount of heat stress experienced by a person include humidity, solar radiation, wind speed, shade, surrounding building materials and physical activity. For example, two walking paths in the same area might expose users to varying levels of heat, even if a general weather app shows the same temperature for both locations.

Thus, this requires a system to observe differences in the environment under street level, and convert complex sensor data into useful guidance. ShadeShift approaches this by building a live heat-risk map of Singapore and making the data available to the public and government agencies.

4 main objectives of the system:

1. Monitor heat and environmental conditions in greater detail than typical weather reporting
2. Predict where and when extreme heat conditions may occur
3. Help pedestrians lower their overall heat exposure by suggesting safer routes
4. Assist authorities in managing cooling facilities and long-term heat mitigation planning

1.3 Real-World Need

While existing public weather information may be useful for general heat conditions, it may not capture the microclimates experienced by pedestrians along specific streets. For example, a walkway exposed to direct sunlight may not be as comfortable compared to a nearby route with sheltered walkways, mature trees or air-conditioned transportation.

Those most affected by this knowledge gap could be elderly residents, small children, those with existing health vulnerabilities, tourists new to the local environment and workers who are outside for extended periods. Those working outdoors, deliverymen, construction workers or event staff could be required to travel or undertake physical activity during periods of High Heat Stress.

ShadeShift aims to provide these users with more practical information instead of simply showing a generic temperature reading. The system could show that one route has more direct sunlight, poor airflow and no nearby cooling facility, but a different route has greater shade coverage and a lower predicted heat-risk score.

The system would also benefit authorities. They may use historical sensor data to determine bus stops, pedestrian crossings and public spaces that experience repeated high heat. This information could help decide where to place additional shelters, trees, drinking-water stations, high-reflectivity materials or automated cooling equipment.

1.4 Target Stakeholders

Stakeholder	Interest in the system
Pedestrian and commuters	Receive heat warnings and receive suggestions on safer walking routes
Elderly and vulnerable users	Reduce exposure to locations with potential high heat exposure
Outdoor workers	
Tourists	Navigate unfamiliar areas while avoiding excessive heat exposure
Event organisers	Monitor outdoor venues and better plan event spaces
Authorities	Identify heat hotspots and place heat-mitigation infrastructure around
Urban planners	Analyse long term heat patterns and test proposed urban improvements

1.5 Existing Systems

Many initiatives already exist which address parts of the urban heat problem. However, they focus on public advisories, environmental planning, or experimental heat-aware route planning rather than integrating all of these capabilities into a single operational system.

1.5.1 NEA Heat Stress Advisory & myENV

Singapore's National Environmental Agency introduced a Heat Stress Advisory using WBGT so that residents can plan prolonged outdoor activities and minimise the risk of heat-related illnesses concurrently. The myENV application also provides environmental information like weather, air quality, flood conditions as well as other services.

These services allow residents to have access to valuable public information. However, ShadeShift helps by focusing on hyperlocal pedestrian exposure. It combines data from street-level sensors, shade information, pedestrian networks and nearby cooling facilities to estimate the heat risk of the particular route they are taking.

1.5.2 Cooling Singapore 2.0

Cooling Singapore 2.0 is developing a Digital Urban Climate Twin that combines environmental, land-surface, industrial, traffic, building-energy and climate models. This helps policymakers evaluate heat-management strategies and better understand Singapore's urban climate.

They also provide a strong foundation for urban-scale research and policy planning. ShadeShift would further improve this by concentrating on live pedestrian services as well as operational responses. It would turn heat information into route recommendations, public alerts, equipment activation and infrastructure maintenance tasks.

1.5.3 Heat-aware route-planning research

Research has shown the likelihood of choosing pedestrian routes based on shade and thermal comfort. The CoolWalks study proposed a route-choice model that takes building shade into consideration while a thermal-comfort path planning project in Austin used a high-resolution thermal modelling to recommend routes with lower heat stress rather than choosing routes based on only distance or travel time.

These studies show that heat aware route planning is feasible. ShadeShift further improves the current approach by integrating live IoT readings, predictive analytics, user alerts and controllable cooling infrastructure within a single smart-city platform.

1.6 Improvements and Innovation

Area	Existing approach	ShadeShift Improvement
Heat information	General weather/ WBGT information	Street level heat risk readings & predictions
Route planning	Based on distance/ travel time	Balance of distance, shade, heat stress and cooling access
Heat response	Users interpret advisories themselves	Personalised alerts & recommend actions
Cooling infrastructure	Managed independently	Connected cooling devices controlled through an IoT platform
Urban planning	Historical studies & simulations	Combination of live operational data with long term analysis

Maintenance	Periodic inspections of public equipment	Sensor-based fault detection & predictive maintenance
Accessibility	General information	Guidance for elderly, outdoor workers and vulnerable groups

ShadeShift is mainly innovated for its closed loop approach. It does not stop collecting and displaying environmental data. The system follows a continuous cycle of:

Sense → Analyse → Predict → Recommend → Respond → Evaluate

Firstly, sensors observe its physical environment. Edge and cloud services then process the data and predict future heat conditions. The system recommends actions to users or authorities, activates cooling facilities around appropriate places and evaluates whether these actions reduce heat risk successfully.

2. Key Functions

2.1 Hyperlocal Heat and Environmental Monitoring

Core function 1: Real-time monitoring of street level heat conditions

IoT sensor stations would be installed at locations like bus stops, pedestrian crossings, parks, bus interchanges, sheltered walkways and outdoor event areas. Depending on location, each station could include the following:

- Air-temperature sensors
- Relative-humidity sensors
- Globe-temperature sensors
- Solar-radiation/ultraviolet sensors
- Wind-speed sensors
- Surface-temperature sensors
- Air-quality sensors
- Rain sensors
- Pedestrian count sensors

These sensor readings would be transmitted to a nearby edge gateway using low-power communication technologies. The gateways would then validate and clean the data, remove invalid readings and calculate heat indicators. Important readings would then be transmitted to the cloud platform for storage, analysis and visualization.

This function is crucial as ShadeShift depends on accurate and timely information on the physical environment. A general city wide temperature reading cannot show the difference in shade, building arrangement, pavement surfaces, ventilation and crowds.

By using a distributed sensor network, the system can provide a more detailed and accurate heat-risk map.

Edge processing also improves reliability. When internet connection is unavailable, the gateway could continue evaluation using local readings and activate an immediate warning when a dangerous threshold is being detected. Once the connection is back, the buffered data could then be uploaded to the cloud.

2.2 AI-based Heat-Risk Prediction and Hotspot Detection

Core function 2: prediction of future heat conditions using artificial intelligence and cloud analytics

The prediction model would process the following:

- Current temperature and humidity readings
- Historical sensor patterns
- Weather forecasts
- Time of day
- Solar exposure
- Building and street geometry
- Vegetation and shade coverage
- Pedestrian density
- Previous heat-related incidents

A time series forecasting model could estimate the likely heat-risk score for each monitored location over the next 30-60 minutes. An anomaly detection model could identify odd temperature changes, malfunctioning sensors or unexpectedly high heat readings.

This function is crucial as a purely reactive system only warns users after conditions have become dangerous. Predictive analysis allows ShadeShift to react at an earlier stage instead. For example, if the system predicts that the walking distance between a MRT station and nearby HDB will experience high heat stress within the next 30 minutes, ShadeShift would then alert commuters through the mobile application and recommend a slightly longer route using sheltered walkways and air-conditioned connections through nearby malls. Cooling fans at selected sheltered areas could also be activated before the predicted heat peak.

The model would run in the cloud as cloud computing provides scalable processing resources and allows data from many sensor stations to be analysed at once. Model outputs would be sent to the mobile application, operations dashboard and digital twin.

2.3 Heat-safe Pedestrian Route Recommendation

Core function 3: route-planning service that reduces a user's expected heat exposure

Traditional route applications usually optimize journeys according to distance of travel time. However, ShadeShift would assign each path segment a dynamic heat-risk weight based on the following:

- Current & predicted WBGT
- Amount of direct sunlight
- Availability of shade
- Surface temperature
- Walking duration
- Crowd density
- Distance from water points
- Availability of indoor/ air-conditioned areas
- Nearby cooling shelters
- Accessibility requirements

The route planning algorithm would calculate several possible paths and suggest an appropriate balance of travel time and heat safety. Users can select their preferences like coolest route, balanced route or shortest route.

This function directly supports the system's goal as cumulative heat exposure depends on both heat intensity and amount of time taken for the route. A slightly longer route may expose a pedestrian to less overall heat than a shorter route under first sunlight.

The route would update dynamically. If a sensor detects a sudden heat increase or a cooling facility becomes unavailable, the cloud service could recalculate the path and notify the user.

2.4 Automated Cooling Response & Infrastructure Management

Core function 4: connection of heat-risk analysis and physical cooling and public warning infrastructure

Possible connected devices include:

- Misting fans
- Smart ventilation fans
- Retractable shades
- Digital information displays
- Warning lights
- Water cooler indicators
- Public address systems
- Smart shutters

When it hits the predicted heat thresholds, the system could activate suitable cooling devices automatically. The response would depend on location, pedestrian presence, available water and energy, weather conditions and operational rules.

For example, a misting fan should not operate continuously when an area is empty or during heavy rain. Pedestrian sensors and rainfall sensors could then be combined with the heat-risk score before the cloud service sends an activation command.

This function is crucial as it turns environmental intelligence into a direct response. It also allows limited cooling resources to be used more efficiently. Instead of blindly operating all devices continuously without precautions, ShadeShift would activate them when and where needed.

Equipment sensors would report information like power consumption, operating temperature, water levels, vibration and device status. AI-based predictive maintenance could also detect patterns that suggest a fan, pump or sensor is likely to fail. Maintenance teams would then receive an alert before a complete breakdown happens.

2.5: Digital Twin for Urban Heat Simulation and Policy Planning

As ShadeShift expands beyond real-time pedestrian routing, it utilises a technology called Digital Twin to support the long term urban planning. A digital twin serves as a virtual replica of Singapore's physical street-level environment, constructed from comprehensive data collection and continuous model training. This core function directly empowers government authorities and urban planners to simulate, analyze, and implement large-scale heat-mitigation infrastructural changes before physical construction begins.

The system utilizes three distinct digital twin types to achieve this comprehensive simulation (Grieves, 2022). First, the Digital Twin Prototype models the foundational physical attributes of urban spaces, encompassing building geometry, street layouts, and existing vegetation. Second, the Digital Twin Instance maps individual, real-time IoT sensor readings, such as global temperature, surface temperature and wind speed onto their specific physical counterparts within the model. Finally, the Digital Twin Aggregate compiles data across all instances to provide a macro-level overview of the city's microclimates and heat patterns.

Through this software application, city planners can perform advanced simulations to test proposed urban improvements. For example, if the system identifies a persistent heat hotspot near a dense commercial district, planners can use the digital twin to virtually introduce new high-reflectivity pavement materials or simulate the growth of

mature shade trees. The digital twin model then recalculates the predicted Wet Bulb Globe Temperature and solar radiation levels to evaluate the efficacy of these changes.

Furthermore, the data collection phase directly feeds into the platform's model training, ensuring the virtual environment accurately mirrors the physical world's thermal dynamics. The continuous evaluation cycle is heavily reliant on this digital twin environment to measure whether automated responses, such as activating misting fans, effectively mitigate heat stress over time. By bridging the gap between live operational data and historical analysis, this function transforms ShadeShift into an essential predictive policy tool, ensuring that limited public resources are deployed effectively and accurately to maximize urban cooling.

3. Dataset Storage

ShadeShift continuously generates and processes heterogeneous data from IoT sensors, pedestrian networks, geographic infrastructure, user reports, and connected cooling devices. A multi-modal data storage approach is therefore suitable because ShadeShift must manage heterogeneous data with different structures and access patterns. Four major data types are handled by ShadeShift: **time-series data, spatial and geospatial data, graph data, and spatial textual data**.

3.1 Streaming Time-Series Data

A major source of data in ShadeShift is the continuous stream of environmental measurements generated by distributed IoT sensor stations. Each reading is associated with a sensor identifier, location, timestamp, and measured value. Examples include air temperature, relative humidity, globe temperature, solar radiation, wind speed, surface temperature, rainfall, pedestrian density, and equipment operating conditions.

These measurements form **time-series data** because their values are recorded sequentially over time (Hu, Sun & Xing 2023). Recent readings are required for real-time heat monitoring, while historical observations are needed to identify temporal patterns and provide input for AI-based heat-risk forecasting. Equipment measurements, such as fan vibration, operating temperature, and power consumption, can similarly be retained over time to support predictive maintenance.

Time-series records can be organised using a timestamp-oriented data model optimised for frequent sequential writes and retrieval over defined time intervals. This is more appropriate than treating every sensor reading as an independent conventional record because ShadeShift may receive measurements continuously from a large number of distributed sensors. Cloud-based storage can also be scaled as additional monitoring stations are deployed, allowing the system to retain both recent operational readings and longer-term historical data.

ShadeShift uses InfluxDB as its time-series database because it is designed for high-volume timestamped data and supports efficient ingestion, retention, and time-range analysis of continuous IoT sensor readings. This makes it suitable for storing both real-time environmental measurements and historical data used for heat-risk forecasting and predictive maintenance.

3.2 Spatial and Geospatial Data

Location is fundamental to ShadeShift because heat exposure varies across the urban environment. The system therefore stores **spatial and geospatial data** representing the coordinates and geometry of sensor stations, pedestrian crossings, bus stops, sheltered walkways, buildings, vegetation, water points, cooling facilities, and identified heat-risk zones.

Spatial information can be represented using points, lines, and polygons. For example, an IoT sensor or drinking-water station can be represented as a point, a pedestrian walkway as a line, and a heat-risk zone or shaded area as a polygon. These representations allow environmental measurements to be associated with their physical surroundings and enable ShadeShift to construct a hyperlocal heat-risk map.

Spatial indexing structures such as **R-trees** can be used to accelerate access to geographically relevant objects without scanning every spatial record (Guttman 1984). Quadrees may also support hierarchical partitioning of geographic space (Samet 1984). These indexes are particularly useful as the number of sensors and connected urban facilities increases. Importantly, R-trees and quadrees function as spatial indexing structures rather than separate database models; they improve the performance of retrieving spatial objects stored within the system.

ShadeShift stores its spatial and geospatial data using PostgreSQL with the PostGIS extension. PostGIS supports geographic objects such as points, lines, and polygons and provides spatial indexing and query capabilities, making it suitable for managing sensor locations, pedestrian paths, heat-risk zones, shaded areas, and cooling facilities.

3.3 Graph Data for Pedestrian Networks

ShadeShift also handles **graph data** to represent the pedestrian transportation network required for heat-safe route recommendation. In this model, intersections, transport entrances, bus stops, building access points, and other important locations can be represented as **nodes**, while pedestrian paths connecting these locations are represented as **edges**.

The pedestrian network is modelled as a weighted graph. Each edge can contain attributes such as walking distance, estimated travel time, shade availability, current and

predicted heat-risk score, pedestrian density, and proximity to cooling facilities. Unlike a conventional route network in which edge weights primarily represent distance or travel time, ShadeShift can dynamically update these attributes as environmental conditions change.

A graph-oriented representation is suitable because route recommendation depends on relationships between connected locations(**Festl et al. 2023**). It enables the system to traverse connected paths and evaluate alternative routes without repeatedly reconstructing complex relationships between separate records. As ShadeShift expands to cover a larger urban area, new pedestrian locations and paths can be incorporated as additional nodes and edges while preserving the existing network structure.

ShadeShift uses Neo4j to store and process the pedestrian network as a property graph. Neo4j allows locations and pedestrian paths to be represented directly as nodes and relationships with associated properties, which supports efficient traversal and dynamic route analysis. This representation is particularly suitable for ShadeShift because route recommendations depend heavily on relationships between connected locations.

3.4 Spatial Textual Data

In addition to quantitative sensor measurements, ShadeShift may receive **spatial textual data** from crowdsourced user reports, maintenance records, and infrastructure incident reports. Such information contains both textual content and geographic context.

For example, an outdoor worker may report through the mobile application that a smart shade at a particular pedestrian crossing is malfunctioning, or a maintenance technician may record an issue with a misting fan at a specific bus interchange. The report contains unstructured or semi-structured text describing the problem, while coordinates or an infrastructure identifier connect the report to a physical location.

These reports can be stored using a document-oriented representation because individual reports may contain different attributes, such as description, timestamp, location, equipment identifier, report category, images or maintenance status. Spatial-textual indexing techniques such as the **IR-tree** can then combine spatial indexing with textual relevance information, supporting efficient retrieval of geographically and semantically relevant reports.

ShadeShift uses Elasticsearch to store and search spatial-textual reports. Elasticsearch supports document-oriented data, full-text search, and geospatial filtering, allowing maintenance teams to search report descriptions while simultaneously restricting results to a relevant geographic area. This makes it suitable for crowdsourced incident reports and cooling-infrastructure maintenance records.

3.5 Integration within the Multi-Modal Data Layer

The four data types are interconnected rather than isolated. A sensor reading has both a timestamp and geographic location; a pedestrian path contains spatial geometry while also belonging to the route graph; and a maintenance report combines textual information with the location and identity of a physical cooling device. ShadeShift therefore benefits from a multi-modal data layer capable of preserving these relationships while using representations appropriate to each type of information.

Data Type	ShadeShift Examples	Suitable Representation	Technology	Primary Purpose
Time-series	Temperature, humidity, WBGT, equipment readings	Timestamp-oriented records	InfluxDB	Real-time monitoring, historical analysis and AI
Spatial/geospatial	Sensors, buildings, shaded areas, cooling facilities	Points, lines and polygons	PostgreSQL + PostGIS	Heat mapping and location analysis
Graph	Intersections and pedestrian pathways	Weighted nodes and edges	Neo4j	Heat-safe route recommendation
Spatial textual	User reports and maintenance records	Document-oriented records	Elasticsearch	Incident search and infrastructure management

By combining these data representations, ShadeShift can store environmental, geographic, network, and textual information in forms appropriate to their structure and intended use. This provides the foundation for the query and analytical operations discussed in the following section.

4. Dataset Query Processing

ShadeShift requires different query and analytical techniques because its core functions depend on temporal, spatial, network, and textual relationships within the collected data. The multi-modal data layer therefore supports **time-series queries**, **spatial queries**, **graph queries**, and **spatial-textual queries**. These queries transform stored data into

information that directly supports hyperlocal monitoring, AI-based heat-risk prediction, heat-safe route recommendation, automated cooling responses, and infrastructure management.

4.1 Time-Series Queries and Analytics

Time-series queries are used to analyse environmental and equipment measurements collected continuously by ShadeShift's IoT sensors. Each measurement contains a timestamp, sensor identifier, location, and measured value, allowing the system to retrieve observations over specific periods and identify changes in environmental conditions.

For example, ShadeShift may execute a query to:

Identify locations where WBGT exceeded a predefined heat-risk threshold during the previous 24 hours.

The system can also calculate statistical summaries such as the minimum, maximum, and average temperature or WBGT for individual monitoring locations over hourly, daily, or weekly intervals. Comparing current observations with historical measurements enables the system to identify recurring periods of high heat exposure and provide historical context for current conditions.

Time-series analytics also support **AI-based heat-risk forecasting**. Recent and historical sequences of temperature, humidity, solar radiation, wind speed, surface temperature, and other environmental variables can be retrieved as input to the forecasting model. Spatio-temporal sensor forecasting is particularly relevant to smart-city applications because urban sensor measurements contain both temporal patterns and relationships between locations (Lu & Li 2024).

Equipment time-series data can similarly be analysed to identify changes in vibration, power consumption, operating temperature, or water level. These results provide input to the predictive-maintenance capability described in the AI section.

4.2 Spatial Queries and Analytics

Spatial queries allow ShadeShift to analyse the geographic relationships between heat conditions, pedestrians, sensors, and cooling infrastructure. Common operations include **range queries, nearest-neighbour queries, point-in-region queries, and spatial proximity analysis**.

For example, when a heat hotspot is detected, the system may perform a range query to:

Identify all sensors and cooling facilities located within 500 metres of the affected area.

A nearest-neighbour query could instead determine the closest drinking-water point, cooling shelter, or air-conditioned public facility to a pedestrian's current location. Spatial

queries can also identify whether a pedestrian route intersects an area currently classified as high heat risk.

These queries directly support the **Hyperlocal Heat and Environmental Monitoring** and **Automated Cooling Response** functions. Instead of treating sensor observations independently, ShadeShift associates environmental conditions with their geographic context. The resulting spatial analysis can be visualised as dynamic heat-risk maps on the mobile application and government dashboard.

Historical spatial analysis can additionally identify persistent heat hotspots by comparing repeated observations across locations and time periods. These patterns can assist urban planners in determining where additional vegetation, shelters, water points, or cooling infrastructure may provide the greatest benefit.

4.3 Graph Queries for Heat-Safe Route Recommendation

Graph queries are central to ShadeShift's **Heat-Safe Pedestrian Route Recommendation** function. The pedestrian network is represented as a weighted graph in which nodes represent intersections, transport entrances, bus stops, and other significant locations, while edges represent walkable connections.

Unlike conventional shortest-path routing, ShadeShift assigns multiple attributes to each edge, including walking distance, estimated travel time, shade availability, pedestrian density, and current or predicted heat risk. These attributes can be converted into a dynamic cost function.

For example:

Find a pedestrian route from an MRT station to an HDB block that minimises cumulative heat exposure while keeping additional walking distance within an acceptable limit. A shortest-path algorithm such as **Dijkstra's algorithm** can then evaluate alternative paths according to the calculated edge costs. Recent heat-aware pedestrian-routing research similarly represents pedestrian networks as routing graphs and modifies edge costs to account for solar exposure. Kolaxidis et al. (2025) found that shade-aware route recommendations could substantially reduce solar exposure with relatively small detours.

This allows ShadeShift to generate the three route options defined in its core functions: **Shortest Route**, **Coolest Route**, and **Balanced Route**. Edge weights can be updated as sensor measurements and heat forecasts change, allowing recommended routes to adapt dynamically rather than remaining fixed throughout the day.

4.4 Spatial-Textual Queries

Spatial-textual queries combine geographic constraints with keywords or textual relevance. They are useful for analysing crowdsourced reports, maintenance records, and incidents associated with specific locations or infrastructure.

For example, an operations officer may request:

Find reports containing “misting fan failure” within one kilometre of the current high-heat-risk zone.

A purely textual search could identify reports containing the relevant terms but would not distinguish incidents located near the affected area from incidents elsewhere in Singapore. Conversely, a purely spatial query could identify nearby reports without determining whether they concern the required type of equipment failure. Combining both conditions produces more operationally useful results.

Spatial-textual queries therefore support the maintenance and infrastructure-management aspects of ShadeShift. They allow government operators to identify geographically relevant equipment problems, group repeated reports concerning the same facility, and prioritise incidents in locations experiencing high heat risk.

4.5 Integration of Queries with ShadeShift Core Functions

The different query types work together to support ShadeShift's major functions rather than operating independently.

Query Type	Example Query	Related ShadeShift Function
Time-series	Identify abnormal WBGT patterns during the previous 24 hours	Heat monitoring and AI heat-risk prediction
Spatial	Find cooling facilities within 500 m of a heat hotspot	Hyperlocal monitoring and automated cooling
Graph	Find the route with the lowest cumulative heat exposure	Heat-safe route recommendation
Spatial-textual	Find nearby reports concerning malfunctioning cooling equipment	Infrastructure monitoring and maintenance

A single operational event may require several query types. For example, when time-series analytics identify an emerging heat-risk hotspot, a spatial query can locate nearby cooling infrastructure, graph processing can update pedestrian route recommendations, and spatial-textual queries can determine whether any relevant equipment has recently been reported as faulty. This integration allows ShadeShift to transform heterogeneous stored data into coordinated real-time decisions.

5. Software Design and Architecture

ShadeShift requires a scalable and fault-tolerant software architecture capable of continuously processing environmental data while supporting real-time heat monitoring, AI-based prediction, pedestrian route recommendation, automated cooling control, and digital-twin services. The proposed system therefore adopts an **event-driven microservices architecture** across IoT device, edge, and cloud layers. Microservices are suitable for large-scale IoT environments because individual system functions can be independently deployed and scaled, while event-driven processing allows services to react to continuously generated sensor data (Khalil et al. 2023).

5.1 Architecture Style

Rather than implementing ShadeShift as a single monolithic application, its major functions are separated into independent cloud microservices. The proposed services include a **Sensor Ingestion Service, Heat Analytics Service, AI Prediction Service, Route Recommendation Service, Cooling Control Service, Notification Service, and Digital Twin Service**.

This separation is important because different ShadeShift functions have different workloads. For example, the Sensor Ingestion Service continuously processes readings from potentially thousands of environmental sensors, whereas the Cooling Control Service is primarily activated when environmental conditions require a physical response. A microservices architecture enables heavily loaded components to scale independently without requiring the entire application to be replicated. Research on IoT architectures similarly identifies independent deployment, scalability, reliability, and availability as important motivations for applying microservices to distributed IoT systems (Khalil et al. 2023).

The architecture is also **event-driven**. Environmental measurements and changes in system state generate events that can be processed by multiple services. For example, when the Heat Analytics Service identifies an increasing WBGT level, the AI Prediction Service can update the future heat-risk forecast, the Route Recommendation Service can recalculate affected routes, and the Notification Service can warn users in the

relevant area. If the risk exceeds a predefined threshold, the Cooling Control Service may also initiate an appropriate cooling response.

This loosely coupled approach allows ShadeShift components to respond to the same environmental event without creating direct dependencies between every service.

5.2 IoT Device, Edge and Cloud Interaction

At the **IoT device layer**, firmware embedded in sensor stations periodically collects measurements including air temperature, relative humidity, globe temperature, solar radiation, wind speed, surface temperature, rainfall, and pedestrian density. Connected actuators such as misting fans, ventilation fans, retractable shades, warning displays, and indicator lights receive commands generated by ShadeShift.

Sensor data are first transmitted to an **edge gateway** rather than sending every raw reading directly to the cloud. The edge gateway validates incoming measurements, filters invalid or duplicate readings, performs basic preprocessing, and calculates preliminary heat indicators. Processing data close to its source can reduce communication latency and unnecessary bandwidth consumption, which is particularly useful for real-time IoT applications (Zhang et al. 2024).

The edge layer also improves resilience. If cloud connectivity is temporarily unavailable, readings can be buffered locally and synchronised when the connection is restored. Predefined safety rules can remain operational at the edge, allowing critical local responses to continue without waiting for the central cloud platform.

Validated sensor events are subsequently transmitted to **Apache Kafka**, which acts as the cloud event-streaming platform and distributes events to the relevant microservices. The broker decouples data producers from consumers so that multiple services can independently process relevant events. The cloud performs computationally intensive operations such as long-term multi-modal data storage, AI inference, historical analytics, heat-risk forecasting, route computation, and digital-twin updates.

Processed information is then delivered to the **mobile application** and **government operations dashboard**. Pedestrians receive heat warnings and route recommendations, while government users can access heat maps, historical analytics, cooling-device status, maintenance alerts, and digital-twin information. Control commands can also travel from the cloud through the edge gateway to physical actuators, producing a bidirectional **physical-to-digital-to-physical feedback loop**.

5.3 Cloud Deployment Strategy

ShadeShift's cloud microservices are deployed using **containerisation**. Each service can be packaged with its runtime environment and software dependencies in a Docker container. Containers provide isolated and portable execution environments, helping services behave consistently across development, testing, and cloud deployment environments.

Containerised services can be managed using **Kubernetes orchestration**. Multiple replicas of heavily used services can be deployed when demand increases, while unhealthy service instances can be replaced automatically. This approach is particularly useful during extreme heat events when large numbers of users may simultaneously request heat-risk information or route recommendations.

A **load balancer** distributes incoming requests across available instances, preventing a single instance from becoming a performance bottleneck. Horizontal scaling can therefore be applied selectively to services experiencing high demand.

ShadeShift can additionally use **serverless functions** for short-lived, event-triggered tasks. For example, a high-heat-risk event could invoke a function that generates location-specific push notifications without requiring a dedicated notification process to remain continuously active. Research on real-time IoT analytics indicates that microservices and Function-as-a-Service approaches can provide modular deployment and scaling capabilities for event-driven workloads (Buzachis et al. 2024).

5.4 Scalability and Fault Tolerance

ShadeShift must remain scalable because its workload will increase as additional sensors, cooling devices, geographic areas, and users are connected. **Horizontal scaling** allows additional instances of high-demand microservices to be created rather than relying solely on increasingly powerful individual servers. Container orchestration and load balancing distribute processing across these instances.

Fault tolerance is implemented at multiple levels. Edge gateways buffer measurements during temporary communication failures, reducing the risk of data loss. In the cloud, service replication prevents the failure of one microservice instance from disabling the entire platform. Health monitoring and automatic service recovery can replace failed instances, while database replication and scheduled backups protect historical and operational information.

The combination of edge and cloud processing further reduces dependence on a single processing location. For example, if connectivity to the cloud is temporarily interrupted during an extreme heat event, the edge gateway can continue evaluating local sensor

measurements and execute predefined safety rules. When connectivity returns, buffered information is synchronised with the cloud for long-term storage and analysis.

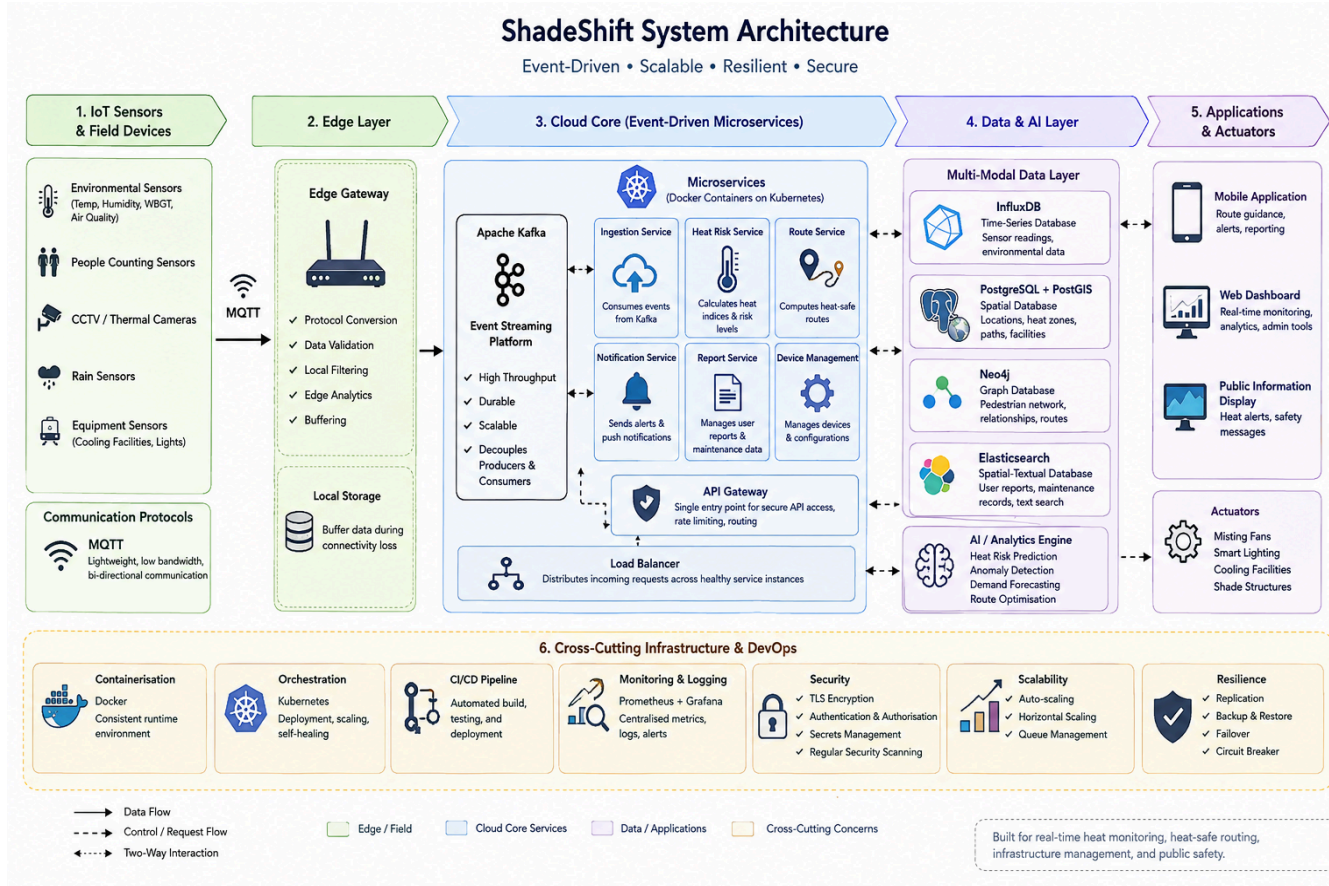
This architecture therefore supports both **scalability and operational resilience**, allowing ShadeShift to expand from selected monitoring locations to a larger smart-city deployment without redesigning the complete software platform.

5.5 ShadeShift Software Architecture

The overall software architecture is illustrated in **Figure X** and follows a bidirectional data flow:

IoT Sensors → Edge Gateway → Cloud Microservices → Multi-modal Data Layer / AI Analytics → Mobile Application, Government Dashboard and Digital Twin

The reverse path allows the Cooling Control Service to send commands through the edge layer to physical actuators. This closed-loop architecture connects environmental sensing, computation, decision-making, and physical response, enabling ShadeShift to move beyond passive monitoring towards active urban heat management.



As illustrated in Figure X, ShadeShift adopts a layered edge-cloud architecture in which environmental data flow from IoT devices through edge gateways and a message broker to cloud-based microservices.

6. Application of AI for Software Engineering

AI can be used to support the development of ShadeShift throughout different stages of the software development process. This is different from the AI heat-risk prediction mentioned earlier in Section 2.2, as this section focuses more on how AI can help the software engineers when building and maintaining the system. Since ShadeShift contains many different microservices, IoT devices, databases and cloud components, AI tools can help reduce some of the repetitive work needed during development. In this project, GitHub Copilot is selected as the primary AI-assisted software engineering tool to support code development, testing and debugging, while all AI-generated suggestions remain subject to human review.

6.1 AI-Assisted Software Development

Generative AI and Large Language Models (LLMs) can assist developers when writing and developing the different components of ShadeShift. Instead of developers having to write every part of the code from scratch, they can describe what they want the program to do and use AI coding tools to generate an initial version of the code.

For example, a developer working on the Sensor Ingestion Service may need a function that receives environmental sensor readings, checks whether the data is valid and then sends it to the message broker. An AI coding assistant could generate a basic API endpoint together with validation and error handling code. The developer can then modify the generated code based on the actual requirements of ShadeShift.

AI can also be used when developing other services such as the Route Recommendation Service, Cooling Control Service and Notification Service. Since these services perform different tasks, using AI may help developers produce basic code structures faster instead of repeatedly creating similar functions manually.

Rasnayaka et al. (2024) studied how LLMs were used in software engineering projects and found that they were useful especially for code generation and debugging. This shows that AI can act as a useful development assistant, however the generated code should still be reviewed by the developer before it is used.

This is important for ShadeShift because some parts of the system can control physical infrastructure. For example, an error in the Cooling Control Service may cause a misting fan or warning system to activate incorrectly. Therefore AI generated code should not simply be directly deployed without being tested and checked first.

GitHub Copilot can support the development of ShadeShift by suggesting code for repetitive and well-defined programming tasks, such as REST API endpoints, sensor-data processing functions, data validation logic and microservice boilerplate code. This can reduce development effort and allow developers to focus on higher-level architectural and system-integration decisions. However, generated code is reviewed and tested by developers before being integrated into the production codebase.

6.2 AI-Assisted Software Testing

AI can also help during software testing. ShadeShift has many different services which interacts with each other, so manually creating every possible test case can become difficult as the system becomes larger.

AI tools can be used to generate test cases based on the software requirements and existing source code. For example, the Heat Analytics Service can be tested to check what happens when WBGT readings go above the heat-risk threshold. Other test cases may include invalid sensor readings, missing data, unavailable cooling devices or temporary network failures.

The Route Recommendation Service could also be tested using different heat conditions, shade coverage and walking distances. For example, a test can check whether the system still recommends the coolest route if one of the usual sheltered walkways becomes unavailable. This is important because ShadeShift's route weights can change depending on current sensor readings and heat predictions.

Tufano et al. (2024) showed that AI-based software development tools are able to generate and run software tests in a controlled development environment. For ShadeShift, similar AI tools could be included within the development pipeline so that tests are automatically performed whenever changes are made to one of the microservices.

However, AI-generated tests should not be the only type of testing used. Developers would still need to perform their own integration, security and system testing. This is especially necessary for cooling equipment because software errors could affect something in the physical environment rather than only causing an application error.

For ShadeShift, GitHub Copilot can assist developers in generating unit-test cases and synthetic test inputs for scenarios such as missing sensor readings, abnormal WBGT values, network interruptions and invalid actuator commands. These AI-generated test cases can supplement manually designed tests and help identify edge cases that may otherwise be overlooked.

6.3 AI-Assisted Debugging and Code Review

Another possible use of AI is debugging. ShadeShift contains many layers of technology, starting from IoT sensors and edge gateways, followed by the message broker, cloud microservices and databases. Because of this, finding the actual source of an error might not always be straightforward.

For example, if temperature readings suddenly stop appearing on the government dashboard, developers may need to check several parts of the system to find out what happened. The problem may come from the sensor itself, the edge gateway, message broker, Sensor Ingestion Service or even the database connection.

An AI debugging tool could analyse error messages, application logs and source code to suggest where the problem may have occurred. This can help developers narrow down the issue faster instead of manually checking every single component. However the developer would still need to confirm whether the suggestion is correct.

AI can also help during code review. For example, it may detect duplicated code, missing error handling, possible programming mistakes or code which does not follow the same coding standards as the rest of the project. Since different developers could be working on separate ShadeShift microservices, this can provide another layer of checking before new code is added into the main system.

During debugging, GitHub Copilot can assist developers by explaining error messages, identifying potential defects and suggesting possible code corrections. In ShadeShift, this may support troubleshooting issues involving sensor-data processing, API communication, database access and event-driven microservice interactions. Suggested fixes are not applied automatically; developers must verify them through code review and testing before deployment.

6.4 Human Oversight when using AI

Although AI can help developers work faster, it should not be allowed to fully develop or deploy ShadeShift by itself. AI-generated code may still contain incorrect logic, security weaknesses or misunderstand the actual requirements of the system.

Human oversight is especially important for ShadeShift because the software is connected to real physical devices. For example, if the Cooling Control Service contains incorrect logic, cooling devices might activate at the wrong time or fail to activate when a high heat risk is detected.

Therefore developers should always review AI-generated code, check suggested bug fixes and verify generated test cases before they are used in the actual system. AI should be treated more as a tool that assists the software engineering team rather than something that replaces the developers.

By combining AI assistance with human review, automated testing and version control, ShadeShift can make the development process more efficient while still making sure the system remains reliable and safe.

Therefore, AI is treated as a development-support tool rather than an autonomous software engineer. Human developers remain responsible for validating generated code, reviewing security implications, conducting system testing and approving changes before deployment.

6.5 AI Models for ShadeShift System Operations

Beyond supporting the software development process, artificial intelligence plays a direct operational role within ShadeShift. Three AI models are integrated into the system to

enable predictive equipment maintenance, real-time sensor anomaly detection, and pedestrian flow analysis. These models run primarily on the cloud platform, with lightweight inference capabilities also deployed at edge gateways for time-critical responses.

6.5.1 Predictive Maintenance Model

ShadeShift operates a distributed network of cooling infrastructure, including misting fans, ventilation units, retractable shade structures and water pumps. Unexpected failures of these devices can leave high-risk pedestrian areas without cooling during heat events, while unnecessary scheduled maintenance wastes operational resources. A predictive maintenance model is therefore deployed to forecast equipment failures before they occur.

The model takes as input time-series features collected from each device, including vibration levels, motor operating temperature, power consumption, water pressure, cumulative runtime and historical maintenance records. An XGBoost gradient-boosted classifier is used to predict the probability of a failure within the next 72 hours, while a regression component estimates the remaining useful life (RUL) of each component. XGBoost is suitable for this task because it handles mixed numerical and categorical features efficiently, provides interpretable feature-importance scores, and performs well on the tabular sensor data collected by ShadeShift (Chen & Guestrin 2016).

When the predicted failure probability exceeds a predefined threshold, the system automatically generates a maintenance alert and assigns it to the relevant technician team through the government operations dashboard. This allows faulty components to be replaced during off-peak periods before they cause service disruption. By shifting from reactive to predictive maintenance, ShadeShift reduces equipment downtime and ensures that cooling infrastructure remains operational when heat risk is highest.

6.5.2 Sensor Anomaly Detection Model

The accuracy of ShadeShift's heat-risk predictions depends entirely on the quality of incoming sensor data. A single faulty temperature or humidity sensor can produce misleading readings that propagate through the Heat Analytics Service and affect route recommendations and cooling control decisions. An anomaly detection model is therefore deployed to identify abnormal sensor readings in real time.

The model uses a two-stage approach. At the edge gateway, an Isolation Forest algorithm performs lightweight anomaly scoring on the raw sensor stream, flagging values that deviate significantly from the expected range based on time of day, location and recent historical patterns. Isolation Forest is efficient for edge deployment because it does not require extensive computational resources and can process high-volume streaming data with low latency (Liu, Ting & Zhou 2008). Suspicious readings are then

forwarded to the cloud, where a deep autoencoder reconstructs the sensor input and flags samples with high reconstruction error as potential anomalies. The autoencoder captures more complex, non-linear patterns that the simpler edge model may miss, such as correlated failures across multiple sensors at the same location.

Detected anomalies are categorised as sensor malfunction, data transmission error, environmental extreme or potential data tampering. The system can then automatically exclude faulty sensors from heat-risk calculations until they are inspected, while notifying maintenance teams of the affected device. This ensures that ShadeShift's decision-making pipelines remain robust even when individual sensors degrade or are compromised.

6.5.3 Pedestrian Flow Computer Vision Model

Understanding pedestrian density is essential for ShadeShift's cooling control and route recommendation functions. Areas with higher pedestrian exposure benefit from prioritised cooling activation, while overcrowded routes may be flagged in favour of less congested alternatives with comparable heat risk. ShadeShift therefore integrates a computer vision model to estimate pedestrian flow and density at monitored locations. The model uses a YOLO (You Only Look Once) object detection architecture to detect and count persons in video frames from selected CCTV cameras at bus interchanges, sheltered walkways and public plazas. YOLO is chosen because it provides a good balance between detection accuracy and inference speed, allowing real-time processing on edge computing hardware (Redmon & Farhadi 2018). A DeepSORT tracking algorithm is applied across consecutive frames to estimate movement direction and avoid double-counting the same individual.

Importantly, the system is designed with privacy preservation as a core requirement. The computer vision pipeline outputs only aggregated counts and density estimates — no facial images, identity information or individual tracking data are stored or transmitted to the cloud. All person-detection processing occurs locally on the edge gateway, and only the numerical count is uploaded. This allows ShadeShift to benefit from crowd-aware cooling optimisation while complying with privacy regulations and avoiding the storage of sensitive biometric data. The resulting pedestrian density estimates are fed into the Cooling Control Service to prioritise device activation in crowded areas and into the Route Recommendation Service to balance thermal comfort against congestion.

7. Use of Networking

ShadeShift relies on a varied mix of IoT devices spread throughout the city. Its networking layer must consistently transfer data from limited, often battery-powered sensors to the cloud. It also needs to send quick and reliable control commands back to physical actuators. This demands careful selection of communication protocols and

network technologies that fit each device type and traffic pattern. A one-size-fits-all networking method won't work across the system.

7.1 IOT Device and Communication Protocols

ShadeShift's device layer can be grouped into three categories with different networking requirements:

Device Category	Examples	Traffic Pattern
Environmental Sensors	Air-temperature, humidity, globe temperature, solar radiation, wind speed, surface temperature, rainfall and air-quality sensors	Small, periodic readings (Every 1-5 minutes)
Vision/Count Sensors	Pedestrian count sensors, cameras at select high traffic nodes	Higher bandwidth, more frequent
Actuators	Misting fans, ventilation fans, retractable shades, warning displays, PA systems, smart shutters	Infrequent but latency sensitive control commands

For sensor-to-gateway communication, ShadeShift uses MQTT, a lightweight publish-subscribe protocol that is suited for devices with limited processing power and intermittent connectivity. Its support for Quality-of-Service levels ensures that critical heat-threshold events are delivered reliably, even over unstable links. The small message overhead helps conserve battery life on distributed stations (Naik 2017). CoAP was considered as an alternative lightweight protocol for highly resource-constrained devices. However, MQTT is selected as ShadeShift's primary IoT messaging protocol because its publish-subscribe model aligns more closely with the system's continuous, event-driven sensor communication. HTTPS is reserved for higher-level, less frequent interactions, such as the mobile application and government dashboard querying the cloud API. In these cases, broader compatibility and simplicity are more important than the extra overhead, since these calls occur less often than continuous sensor telemetry.

This protocol selection reflects the different communication requirements within ShadeShift. MQTT is used for continuous IoT telemetry because its lightweight publish-subscribe model reduces communication overhead for constrained devices, while HTTPS is used for less frequent, higher-level application traffic where secure request-response communication and broad compatibility are more important (Naik 2017).

7.2 Network Technologies

Because ShadeShift's environmental sensor stations are distributed across streets, parks, bus stops and walkways, many of which may have limited access to mains power, the system uses LoRaWAN for low-power sensor connectivity. LoRaWAN is designed for long-range, low-data-rate communication and is therefore suitable for battery- or solar-powered environmental sensors that periodically transmit small measurements. Its ability to support large numbers of devices through gateway-based connectivity also makes it suitable for scaling ShadeShift across a large urban area (Ogbodo, Abu-Mahfouz & Kurien 2022). NB-IoT was considered as an alternative LPWAN technology; however, LoRaWAN was selected for the proposed architecture because ShadeShift primarily requires low-power transmission of small, periodic environmental measurements.

Actuators and vision-based sensors have different needs. Cooling devices require faster and more reliable command delivery compared to periodic sensor uploads, while cameras create larger data packets. Therefore, these devices connect via Wi-Fi where there is fixed power and local infrastructure available, such as in bus interchanges and sheltered walkways. For the edge-gateway-to-cloud backhaul link, 5G is used because its high bandwidth and low-latency communication can support aggregated sensor events, processed pedestrian-count data and time-sensitive control acknowledgements between the edge layer and cloud services.

This hybrid networking architecture uses LoRaWAN for low-power environmental sensing, Wi-Fi for higher-bandwidth local devices and actuator connectivity, and 5G for edge-to-cloud backhaul. Each technology is therefore matched to the bandwidth, latency, coverage and energy requirements of its respective device category, rather than forcing all components to rely on a single network technology (Ogbodo, Abu-Mahfouz & Kurien 2022).

8. Advanced Technologies and Security

Beyond its main sensing and cooling functions, ShadeShift includes digital twin, blockchain, and cybersecurity features. These support long-term urban planning, infrastructure accountability, and protection against the increased risks from a city-scale IoT deployment.

8.1 Digital Twin Integration

As described in Section 2.5, ShadeShift's Digital Twin Instance layer requires continuous, bidirectional synchronisation between physical sensors/actuators and their virtual counterparts, so that planners can simulate proposed heat-mitigation changes

against an accurate live model rather than a stale snapshot. This depends directly on the networking infrastructure described in Section 7: the 5G backhaul and Apache Kafka event-streaming architecture described in Section 5.2 must support low-latency and highly available data exchange to keep the digital twin synchronised with real-world conditions.

Using this continuously updated virtual environment, urban planners can evaluate heat-mitigation strategies before physical deployment. For example, the digital twin can simulate the potential effects of additional shade structures, vegetation or cooling facilities on pedestrian heat exposure. This allows interventions to be compared under different environmental scenarios before resources are committed to real-world implementation.

8.2 Blockchain

ShadeShift applies blockchain technology to two areas where trust and auditability are important: infrastructure accountability and automated maintenance coordination. Records of cooling device activations, such as a misting fan being triggered in response to a predicted heat spike, can be written to a permissioned blockchain ledger shared between the operating authority and any external contractors managing cooling equipment. Because blockchain records are cryptographically linked and effectively tamper resistant, this provides an auditable trail proving that public cooling infrastructure was activated appropriately and on time, which is useful both for public accountability and for evaluating whether interventions actually reduced heat exposure. Blockchain based frameworks combining decentralised ledgers with AI-driven anomaly detection have been shown to improve data integrity and reduce trust related vulnerabilities in smart city IoT deployment (Islam et al. 2025).

8.3 Smart Contracts

Smart contracts extend this further by automating parts of the Automated Cooling Response and Infrastructure Management function(Section 2.4). For example, when an equipment sensor confirms that a faulty misting fan has been repaired and passes a diagnostic check, a smart contract could automatically release payment to the maintenance contractor without manual invoicing, reducing administrative delay and disputes. This mechanism must be applied carefully. Smart contracts execute automatically according to the data supplied to them, but they cannot independently verify whether a real-world event, such as a completed repair, has genuinely occurred. Instead, they rely on external data sources, commonly referred to as oracles. This creates the oracle problem: if a sensor is faulty, delayed or compromised, the smart contract may execute correctly according to its programmed logic but on the basis of

incorrect information. ShadeShift mitigates this risk by requiring multi-source confirmation before a smart contract is triggered. For example, completion of a repair can be verified using both an equipment diagnostic result and a technician's authenticated digital sign-off rather than relying on a single IoT sensor.

8.4 Defense Against Emerging Cyber Attacks

A distributed sensor and actuator network of ShadeShift's scale is a meaningful cyber attack surface. IoT devices are frequently targeted because many run on constrained hardware with weak default security, making them attractive targets for botnet-based DDoS attacks, where large numbers of compromised devices are hijacked to flood a target with traffic and disrupt availability (Kim et al. 2023). Given that ShadeShift's cooling actuators and warning systems support public safety functions, such disruption could have real physical consequences during an active heat event, not just a data outage. Other relevant threats include man-in-the-middle attacks against device communications, unauthorised actuator control, credential theft, sensor-data manipulation and malicious firmware modification.

ShadeShift addresses this risk through layered defences aligned with the architecture in Section 5:

- Device-level authentication — each device uses unique credentials and TLS-encrypted communication rather than shared or default passwords, reducing the risk of unauthorised device access.
- Network segmentation - the sensor/actuator network is isolated from the public facing mobile application and dashboard behind a DMZ, so a compromise of one does not automatically expose the other.
- Edge-level validation - the edge gateways described in Section 5.2 flag traffic patterns consistent with device compromise, such as abnormal transmission frequency, in addition to filtering invalid readings.
- Regular firmware updates - sensor and gateway firmware is updated over the air on a scheduled basis to patch known vulnerabilities, since outdated firmware is one of the most common reasons IoT devices are successfully exploited.
- Cloud-level intrusion detection — the cloud platform monitors anomalous traffic patterns and traffic volumes associated with potential DDoS activity, allowing suspicious traffic to be identified and mitigated before the availability of critical actuators or user-facing applications is significantly affected.
- Role-based access control (RBAC) — access to administrative dashboards, device-management functions and actuator-control services is restricted according to authorised user roles, reducing the risk of unauthorised changes to critical infrastructure.

Together, device authentication, encrypted communication, network segmentation, edge-level validation, secure firmware maintenance, role-based access control and cloud-level intrusion detection provide a defence-in-depth security strategy for ShadeShift. These controls reduce the likelihood that compromise of a single device or network component can propagate across the wider city-scale IoT system.

References

- Al-Doghman, F, Moustafa, N, Khalil, I, Sohrabi, N, Tari, Z & Zomaya, AY 2023, 'AI-enabled secure microservices in edge computing: Opportunities and challenges', *IEEE Transactions on Services Computing*, vol. 16, no. 2, pp. 1485–1504, doi:10.1109/TSC.2022.3155447.
- Buo, I, Khan, WH, Crabtree, E, Emmott, F, Hariyani, D & Middel, A 2026, 'Cool routes: Real-time human thermal exposure routing', *Building and Environment*, vol. 298, article 114622, doi:10.1016/j.buildenv.2026.114622.
- Festl, K, Promitzer, P, Watzenig, D & Yin, H 2023, 'Performance of graph database management systems as route planning solutions for different data and usage characteristics', *2023 IEEE International Automated Vehicle Validation Conference (IAVVC)*, doi:10.1109/IAVVC57316.2023.10328126.
- Guttman, A 1984, 'R-trees: A dynamic index structure for spatial searching', *ACM SIGMOD Record*, vol. 14, no. 2, pp. 47–57, doi:10.1145/971697.602266.
- Hu, C, Sun, Z, Li, C, Zhang, Y & Xing, C 2023, 'Survey of time series data generation in IoT', *Sensors*, vol. 23, no. 15, article 6976, doi:10.3390/s23156976.
- Islam, R, Bose, R, Roy, S, Khan, AA, Sutradhar, S, Das, S, Ali, F & AlZubi, AA 2025, 'Decentralized trust framework for smart cities: A blockchain-enabled cybersecurity and data integrity model', *Scientific Reports*, vol. 15, article 23454, doi:10.1038/s41598-025-06405-y.
- Kim, D, Jeon, S, Shin, J & Seo, JT 2023, 'Design the IoT botnet defense process for cybersecurity in smart city', *Intelligent Automation & Soft Computing*, vol. 37, no. 3, pp. 2979–2997, doi:10.32604/iasc.2023.040019.
- Kolaxidis, N, Ludwig, C, Knoblauch, S, Fürle, J, Foshag, K, Fendrich, S, Lautenbach, S & Zipf, A 2025, 'Mitigating heat stress by reducing solar exposure in pedestrian routing', *Transactions in GIS*, vol. 29, no. 6, article e70110, doi:10.1111/tgis.70110.
- Li, S, Lai, L, Wang, C, Yao, C, Wang, J, Wang, J, Mao, H & Santamouris, M 2026, 'Evaporative cooling mist-spray system to enhance thermal comfort in urban outdoor premises', *Energy and Buildings*, vol. 352, article 116834, doi:10.1016/j.enbuild.2025.116834.
- Lu, J & Holubová, I 2019, 'Multi-model databases: A new journey to handle the variety of data', *ACM Computing Surveys*, vol. 52, no. 3, article 55, pp. 1–38, doi:10.1145/3323214.

Lu, Y-J & Li, C-T 2024, 'Forecasting urban sensory values through learning attention-adjusted graph spatio-temporal networks', *ACM Transactions on Spatial Algorithms and Systems*, vol. 10, no. 1, article 4, pp. 1–22, doi:10.1145/3635140.

Naik, N 2017, 'Choice of effective messaging protocols for IoT systems: MQTT, CoAP, AMQP and HTTP', *2017 IEEE International Systems Engineering Symposium (ISSE)*, pp. 1–7, doi:10.1109/SysEng.2017.8088251.

National Environment Agency (NEA) 2025, *2024 is Singapore's warmest year on record*, National Environment Agency, 23 March, viewed 16 August 2026, <https://www.nea.gov.sg/media/news/news/index/2024-is-singapore-s-warmest-year-on-record>.

National Environment Agency (NEA) & Ministry of Sustainability and the Environment (MSE) 2023, *New heat stress advisory launched to guide public on minimising risk of heat-related illnesses*, National Environment Agency, 24 July, viewed 16 August 2026, <https://www.nea.gov.sg/media/news/news/index/new-heat-stress-advisory-launched-to-guide-public-on-minimising-risk-of-heat-related-illnesses>.

Ogbodo, EU, Abu-Mahfouz, AM & Kurien, AM 2022, 'A survey on 5G and LPWAN-IoT for improved smart cities and remote area applications: From the aspect of architecture and security', *Sensors*, vol. 22, no. 16, article 6313, doi:10.3390/s22166313.

Rasnayaka, S, Wang, G, Shariffdeen, R & Iyer, GN 2024, 'An empirical study on usage and perceptions of LLMs in a software engineering project', *arXiv preprint*, arXiv:2401.16186, doi:10.48550/arXiv.2401.16186.

Samet, H 1984, 'The quadtree and related hierarchical data structures', *ACM Computing Surveys*, vol. 16, no. 2, pp. 187–260, doi:10.1145/356924.356930.

SMU City Perspectives Team 2022, 'Cooling Singapore 2.0: A step towards becoming a climate resilient and regenerative city', *City Perspectives*, Singapore Management University, 8 September, viewed 16 August 2026, <https://cityperspectives.smu.edu.sg/article/cooling-singapore-20-step-towards-becoming-climate-resilient-and-regenerative-city>.

Tufano, M, Agarwal, A, Jang, J, Zilouchian Moghaddam, R & Sundaresan, N 2024, 'AutoDev: Automated AI-driven development', *arXiv preprint*, arXiv:2403.08299, doi:10.48550/arXiv.2403.08299.

Chen, T & Guestrin, C 2016, 'XGBoost: A scalable tree boosting system', *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794, doi:10.1145/2939672.2939785.

Liu, FT, Ting, KM & Zhou, Z-H 2008, 'Isolation forest', *2008 Eighth IEEE International Conference on Data Mining*, pp. 413–422, doi:10.1109/ICDM.2008.17.

Redmon, J & Farhadi, A 2018, 'YOLOv3: An incremental improvement', arXiv preprint, arXiv:1804.02767.

Tusa, F, Clayman, S, Buzachis, A & Fazio, M 2024, 'Microservices and serverless functions—lifecycle, performance, and resource utilisation of edge based real-time IoT analytics', *Future Generation Computer Systems*, vol. 155, pp. 204–218, doi:10.1016/j.future.2024.02.006.